

Student Dropout Prediction

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Abstract—This research uses a dataset of 649 students with academic, social, and demographic data to demonstrate a machine learning method for predicting student dropout. Encoding, scaling, stratified splitting, and class imbalance management with SMOTE were all part of the data preprocessing. The best indicators of dropout were found to be academic performance, especially grades and past failures. Using metrics appropriate for unbalanced data, such as F1-score and recall, several models were examined including Logistic Regression, Decision Tree, Random Forest, SVM, KNN, Naive Bayes, Gradient Boosting, and XGBoost. The best performance is achieved by tree-based and ensemble models, according to the results. Cross-validation, however, suggests that the short dataset size may cause overfitting. These results underscore the importance of model selection and feature engineering while emphasizing the need for better generalization in future studies.

I. INTRODUCTION

A. Research Problem and prediction Target

Student dropout in higher education is a topic of interest because it's a significant problem. Students who choose to drop out face long-term consequences, whether academically or professionally. On the other hand, academic institutions such as universities are also negatively affected by the reduced graduation rates and the wasted resources. The research problem addressed in this project is the accurate early prediction of a student's possibility of dropping out at the beginning of their academic journey. This task is a supervised learning problem in which the goal is to predict whether a student will drop out or continue their academic journey. This prediction could be modeled as a binary classification problem (dropout vs. non-dropout).

B. Motivation

The completion of this predictive model will enable early identification of dropout risk, a proactive measure that universities can integrate into their systems to provide accurate guidance, such as academic advising, financial support, mental health support, or intensive tutoring. Manual evaluation of at-risk students is solely based on indicators that are usually recognized late. The implementation of machine learning techniques helps us analyze large amounts of data to detect early signs of dropout. The application of predictive models can help students address the factors affecting them and driving them to drop out early, before the problem persists and harms their academic performance. This predictive model will help students' retention and increase their academic success.

C. Problem Formulation

This study uses a publicly accessible higher education dataset that includes demographic, socioeconomic, and early academic performance variables to construct a supervised machine learning classification model for student dropout prediction. The goal is to develop a prediction model that outputs the student's dropout risk based on a set of input characteristics describing the student's background and

academic achievement. To ensure accurate evaluation of model performance, suitable evaluation metrics, including F1-score, recall, are considered, given the inherent class imbalance in dropout datasets.

D. Intended Contributions

This attempt aims to make three contributions. Using a reputable benchmark dataset, it first offers a methodical machine learning approach for early prediction of student dropout. Second, to find efficient methods for this task, it assesses and contrasts a range of baseline and cutting-edge machine learning models. Lastly, it supports understandable, useful uses of predictive analytics in higher education by providing insights into the major determinants of student dropout.

II. BENCHMARKS

- F1-Score: The mean of precision and recall (primary for imbalanced data)
- Recall/Sensitivity (critical for not missing at-risk students)
- Precision (to avoid over-identifying false positives)
- G-Mean (geometric mean of sensitivity & specificity)

III. DATA SET

- <https://www.kaggle.com/datasets/abdullah0a/student-dropout-analysis-and-prediction-dataset>
- The dataset is explicitly designed for analyzing factors that influence student dropout and building predictive models for dropout risk.
- Explore which features (e.g., demographics, academic performance) correlate with dropout outcomes.
- The data set has 34 columns (features) and 700 rows (students)

IV. RELATED WORK

5 papers were studied to establish a standard for our student dropout prediction model. The papers are published within the last 3 years to ensure the integration of the latest technologies for optimal model performance. The papers discussed and emphasized the importance of selecting a strong model, feature selection, and dropout reasoning.

A. All-Year Dropout Prediction Modeling and Analysis for University Students (2023)

a) Task Definition

This paper aims to determine the best format for students' data to use in modeling the prediction system, without considering the students' current year of study. The research paper argues that different levels in their studies entail distinct features, so the researchers aimed to develop and use only the most important information. The research paper tested many different approaches and data to possibly find the best way to represent the data

b) Models, Data, & Features Used

Models: the study consisted of six models: Decision Tree, Random Forest, LightGBM, XGBoost, Logistic Regression, and Support Vector Machine.

Data: data of 60010 students were collected from an unspecified university in Korea from 2010 to 2021. Of the total 60010 students, only 29099 have graduated, 6963 have dropped out of school, and the remaining 23948 are currently in school or suspended.

Features: 23 features categorized into 4 classes: students' grades, students' attendance, students' scholarships, and students' personal backgrounds.

c) Evaluation Metrics & Experimental Setup

Metrics: There were 4 success metrics: accuracy, precision, recall, and F1 score.

Experimental setup: the data was split into 70% for training and 30% were used to test the model, and there was also a prediction set. Then, each model was evaluated on each feature table to assess its performance in detail.

d) Limitations & Research Gaps

In the All-Year Dropout Prediction Modeling and Analysis for University Students, the model was limited to this Korean university and was not generalizable to other universities. The paper may also reflect some bias, as the study was conducted at a private university and does not generally generalize well to public universities. The research gaps identified were that this paper typically focuses on financial and demographic features rather than the academic features that other papers do.

B. Student Dropout Prediction for University with High Precision and Recall (2023)

a) Task Definition

The purpose of this paper is to propose a student dropout prediction system to identify students who are about to drop out, with the aim of also categorizing the dropout reasons to meet the needs of the offices of academic affairs. The consulting institute needs an accurate prediction model. The model is designed to aim for high dropout precision and high dropout recall rate with the integrated representation of dropout prediction and categorized reasoning into four categories such as “employed”, “Did Not Register”, “Personal Issue”, and “Admitted to Other University.” helps in a customized advising session to prevent the percentage of dropouts each year increasing the efficiency of counseling/advising sessions.

b) Models, Data, & Features Used

Models: the student dropout prediction system used XGBoost with SMOTE (to ensure high precision) and CatBoost with RandomOverSamplerSMOTEEN (to ensure high recall). The model then combined the prediction results in a combiner to predict non-dropout, high-risk, or low-risk. The prediction model prioritized higher recall over a higher priority.

Data: The researchers collected student records from Gyeongsang National University from 2016 to 2022. Then, data preprocessing was applied to finally get a total of 67,060 students. Over the 5 years during which the data were collected, 5.1% of the students dropped out. The key trend was that male students were more likely to drop out than female students. It was also noted that freshmen are more likely to drop out than seniors, highlighting the importance of intensive counseling during the first years. Other factors were

studied, like the reason for dropout. Most notable was being admitted to other universities; after that, students had a hard time, whether due to health or family issues.

Features: 27 features divided into 5 feature classes, which are academic data, academic records, personal information, facility use history, and website use history. Permutation importance and SHAP analysis were used to reduce the feature set and improve prediction performance.

c) Evaluation Metrics & Experimental Setup

Metrics:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$DropoutRecall = \frac{TP}{TP + FN}$$

$$DropoutPrecision = \frac{TP}{TP + FP}$$

$$DropoutF1 = 2 * \frac{DropoutPrecision * DropoutRecall}{DropoutPrecision + DropoutRecall}$$

Experimental setup: For dropout prediction, 70% of the data was used to train the model, 30% for testing, and 10-fold cross-validation was also used. To categorize student dropout reasons, 80% of the data was used for training and 20% for testing. A 10-fold cross-validation was also used, and K-means clustering and PCA were used.

d) Limitations & Research Gaps

In the Student Dropout Prediction for University with High Precision and Recall paper they had a few limitations the study was performed only one university the training and testing data was also performed on the Gyeongsang National University making this model not generalizable, also the model was trained on data that heavily relied on the university system such as website login count which may not be widely available for many universities. The research gaps mentioned were that other models prioritize one metric over another, and that other models don't categorize the reasons for dropout.

C. Predicting Student Dropout Rates Using Supervised Machine Learning (2024)

a) Task Definition

The prediction of student dropouts as a supervised machine learning problem is studied in this work. Using demographic, socioeconomic, and educational data gathered from a broad education survey, the objective is to estimate whether a student will complete their education or drop out. Finding the main causes of dropout and encouraging early intervention techniques to increase student retention are the aims.

b) Models, Data, & Features Used

The study covers a number of supervised learning models, such as K-Nearest Neighbors, Random Forest, Naïve Bayes, Logistic Regression, Probit Regression, Decision Tree, and Support Vector Machine. The National Education Accessibility Survey (NEAS) carried out in Somaliland in 2022 is the origin of the data. The feature set consists of socioeconomic factors like household income and housing type, demographic variables like age and grade level, and educational environment elements like school type. The binary target variable indicates whether a person is a dropout or not.

c) Evaluation Metrics & Experimental Setup

Accuracy is the main metric used to compare the overall prediction power of different classifiers. Additionally, ROC-AUC is utilized to evaluate the models' ability for discrimination, and F1-score is presented to adjust for possible class imbalance in the dropout data. Preprocessing the survey data, training several classifiers on the complete feature set, and evaluating their predicted performance form the experimental design. Random Forest has the best overall accuracy of all the models that were evaluated.

d) Limitations & Research Gaps

This study's reliance on a Somaliland-specific dataset is one of its main limitations, which may restrict the results' applicability to other educational systems or geographical settings. Additionally, the study gives little attention to advanced imbalance-handling strategies or in-depth recall-based analysis, which are crucial for identifying at-risk students, even though accuracy and F1-score are provided. Additionally, the experimental design reduces repeatability and benchmarking clarity by offering scant information on cross-validation and hyperparameter adjustment.

D. Enhancing the Early Student Dropout Prediction Model Through Clustering Analysis of Students' Digital Traces (2024)

a) Task Definition

The goal of this work is to enhance the prediction of early student dropouts by combining conventional machine learning classification with clustering analysis of digital interaction data. The assignment is presented as a supervised learning problem in which the target variable indicates whether a student will drop out, and instances represent specific students. By first classifying students with similar behavioral profiles based on their digital traces and then using classification models within these clusters to identify latent patterns that affect dropout probability, the main goal is to increase predictive accuracy.

b) Models, Data, & Features Used

In order to predict dropout risk, the study used a two-stage modeling technique that first groups students based on similarities in their behavioral data using unsupervised clustering. The resulting clusters are then subjected to supervised classification models, such as decision trees and ensemble learners. Instead of using traditional academic records, the data is derived from students' digital traces on educational platforms, such as learning management system logs, which record patterns of online activity. The behavior-based characteristics, such as interaction counts, usage frequency, access duration, and activity patterns, enable clustering to uncover significant engagement profiles that improve the classifiers' prediction capabilities.

c) Evaluation Metrics & Experimental Setup

Standard classification metrics suitable for imbalanced datasets are used to evaluate the performance of the improved dropout prediction model. These metrics include precision and ROC-AUC to measure overall classification quality and discrimination across thresholds, recall to evaluate the model's ability to identify at-risk students, and F1-score to balance precision and recall between dropout and non-dropout classes. The experimental approach involves preprocessing students' digital traces, applying clustering to group behaviorally similar students, and training classification models on the clustered data, with results

evaluated to baseline models without clustering to demonstrate performance increases.

d) Limitations & Research Gaps

The method has a number of drawbacks, such as its reliance on comprehensive digital interaction logs that might not be uniformly accessible or standardized across educational institutions, the sensitivity of the clustering stage to algorithm selection and parameter settings (like the number of clusters), which can impact reproducibility and generalizability, and limited external validity because the behavior-based features are derived from a particular institutional context and might not transfer well to datasets that are dominated by academic or demographic data.

E. A Study on Dropout Prediction for University Students Using Machine Learning (2023)

a) Task Definition

The study addresses the supervised machine learning classification problem of student dropout prediction. The goal is to develop and evaluate predictive models that use academic records and performance history to forecast a university student's likelihood of dropping out. Accurate early prediction is intended to direct educational interventions. Dropout is defined as students who leave their program before finishing it.

b) Models, Data, & Features Used

In order to address class imbalance, the authors used oversampling techniques like SMOTE, ADASYN, and Borderline-SMOTE while evaluating a number of machine learning models, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, Deep Neural Network, and Light Gradient Boosting Machine. The dataset, which focuses on academic-related characteristics, comprises 20,050 university student records gathered from an institutional database in Korea. Measures of academic achievement and student development, such as cumulative GPA, enrollment history, credits gained, and other indicators pertinent to forecasting student dropout risk, are included in the features.

c) Evaluation Metrics & Experimental Setup

The performance of models was evaluated using F1-score as the primary metric due to the imbalanced nature of dropout instances. To address class imbalance, models were trained and tested on the entire dataset both with and without oversampling. LightGBM demonstrated the efficacy of gradient boosting with imbalance handling by achieving the highest F1-score (0.840). In order to determine how imbalance correction affects predictive accuracy, the experimental setting compares baseline algorithms with oversampled variations.

d) Limitations & Research Gaps

A thorough examination of model performance is limited by the study's major focus on F1-score rather than other evaluation measures like recall, precision, and ROC-AUC. Additionally, the employment of deep learning and ensemble models presents interpretability issues for educational stakeholders, and it primarily relies on oversampling approaches that may generate artificial data artifacts and limit generalizability. Furthermore, the findings may not be as applicable to colleges with various academic environments or student populations due to the dependence on data from a single university.

V. SUMMARY TABLE OF PRIOR WORK

TABLE I

Paper	Method	Dataset	Evaluation Metrics
All-Year Dropout Prediction Modeling and Analysis for University Students	- Decision Tree - Random Forest - LightGBM - XGBoost - Logistic Regression - Support Vector Machine	- 60010 students - Collected from an unspecified university in Korea - From 2010 to 2021 - 23 features (divided into 4 categories)	- Accuracy - Precision - Recall - F1 score
Student Dropout Prediction for University with High Precision and Recall	Hybrid model combining: 1. XGBoost + SMOTE 2. CatBoost + RandomOverSamplerSMOTEENN 3. PCA + K-means clustering for dropout reason categorization	- 67,060 student record - Gyeongsang National University - From 2016 to 2022 - 27 features (divided into 5 categories)	- Accuracy - Dropout Recall - Dropout Precision - Dropout F1 - 7:3 ratio for dropout prediction - 8:2 ratio for categorization of reasoning
Predicting Student Dropout Rates Using Supervised Machine Learning	Predicting student dropout using supervised ML	NEAS 2022 (demographic & educational survey data, Somaliland)	Accuracy, F1-score, ROC-AUC
Enhancing the Early Student Dropout Prediction Model Through Clustering Analysis of Students' Digital Traces	Early student dropout prediction with clustering	Digital traces(LMS interactions)	F1-score, Recall, Precision, and ROC-AUC
A Study on Dropout Prediction for University Students Using Machine Learning	University student dropout prediction	20,050 student academic records	F1-score

VI. DEFINED BENCHMARK METRICS FOR EXPERIMENTS

- Overall Accuracy: Provides a general performance understanding
Primary Metric:
- F1-Score: Balances precision and recall (harmonic mean) for imbalanced classification.
Key Intervention Metrics:
- Recall (Sensitivity): The most critical metric is the ability to identify at-risk students.
- Precision: Ensures prediction reliability for efficient resource use (minimizing resource waste).
Overall Performance Metrics:
- ROC-AUC: Measures the model's ability to separate classes across thresholds.
- G-Mean: Guarantee balanced performance across dropout and non-dropout classes.
Supplementary Metric:
- Accuracy: Provides general performance understanding, but is not prioritized due to class imbalance.

VII. DATASET CHARACTERIZATION, PREPROCESSING, AND EDA

A. Dataset Description

The dataset used in this project is the Student Dropout Analysis and Prediction Dataset, which was obtained from Kaggle. It is a tabular dataset that was originally collected from a secondary school environment and contains demographic, social, and academic features for each student. The dataset has 649 students (rows) and 34 columns, where 33 are input features and 1 is the target variable (Dropped_Out).

TABLE II

Property	Detail
Dataset Source	Kaggle (abdullah0a)
Data Type	Tabular / Mixed (Numerical + Categorical)
Total Instances	649 Students
Total Features	33 features + 1 target column
Target Variable	Dropped_Out (Boolean -> mapped to 0/1)
Missing Values	None
Class Distribution	Not Dropped: 549 (84.6%) Dropped: 100 (15.4%)

The features can be grouped into four categories: demographic features (e.g., age, gender, address), family background features (e.g., parent education, parent job, guardian), behavioral and social features (e.g., study time, going out, alcohol consumption, absences), and academic performance features (e.g., Grade_1, Grade_2, Final_Grade, number of failures). The target variable was originally stored as a Python boolean and was converted to integer format (0 = did not drop out, 1 = dropped out) before any processing.

B. Preprocessing Steps

a) Missing Values

After loading the dataset, we checked all 34 columns for null values. No missing values were found in any column, so no imputation was needed.

b) Categorical Encoding

Several columns were stored as string categories, including School, Gender, Address, Family_Size, Parental_Status, Mother_Job, Father_Job, Reason_for_Choosing_School, Guardian, School_Support, Family_Support, Extra_Paid_Class, Extra_Curricular_Activities, Attended_Nursery, Wants_Higher_Education, Internet_Access, and In_Relationship. A total of 17 columns were label-encoded using scikit-learn's LabelEncoder, converting each unique string value to a corresponding integer.

c) Feature Scaling

After encoding, all 37 features (including the 4 engineered ones described in Section 2) were scaled using StandardScaler to bring them to zero mean and unit variance. This is important for models like Logistic Regression, SVM, and KNN which are sensitive to feature magnitude differences.

d) Train/Test Split

The dataset was split into 80% training (519 students) and 20% test (130 students) using stratified splitting to maintain the same class ratio in both sets. The dropout rate in both splits was 15.4%, confirming the stratification worked correctly.

e) Handling Class Imbalance with SMOTE

Since only 15.4% of students are labeled as dropped out, the dataset is significantly imbalanced. To deal with this, we applied SMOTE (Synthetic Minority Oversampling Technique) on the training set only. After SMOTE, the training set grew from 519 to 878 samples with an equal split of 439 dropout and 439 non-dropout instances. The test set was left untouched to keep evaluation realistic.

C. Justification of Evaluation Protocol

Given the class imbalance in this dataset, accuracy alone would be misleading. For example, a model that predicts every student as non-dropout would achieve 84.6% accuracy without identifying a single at-risk student. For this reason, the following metrics were selected based on our Phase 1 benchmark definitions:

- F1-Score – primary metric; balances precision and recall for the imbalanced dropout class
- Recall – most critical metric for this problem; we do not want to miss students who are at risk
- Precision – ensures we are not flagging too many students incorrectly
- G-Mean – geometric mean of sensitivity and specificity; ensures balanced performance
- Accuracy – included for reference but not used as a primary metric.

A 5-fold stratified cross-validation was also performed on the

best model to verify that results are stable and not just due to a lucky train/test split.

D. Exploratory Data Analysis (EDA)

a) Class Distribution

The class distribution is heavily skewed. Out of 649 students, 549 (84.6%) did not drop out and only 100 (15.4%) dropped out. This confirms what was mentioned in prior studies – dropout datasets are typically imbalanced and require special handling during training.

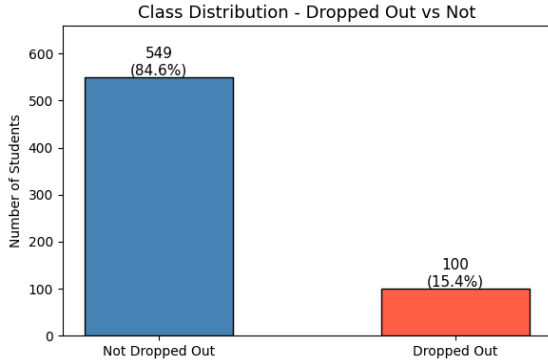


Figure 1: Class distribution showing significant imbalance between dropout and non-dropout students

b) Feature Distributions by Class

Looking at the distributions of key features split by dropout status, there are clear differences. Dropout students tend to have lower grades across all three assessments, more failures, and higher absenteeism compared to students who stayed enrolled. The study time distribution also shows that students who dropped out generally studied less.

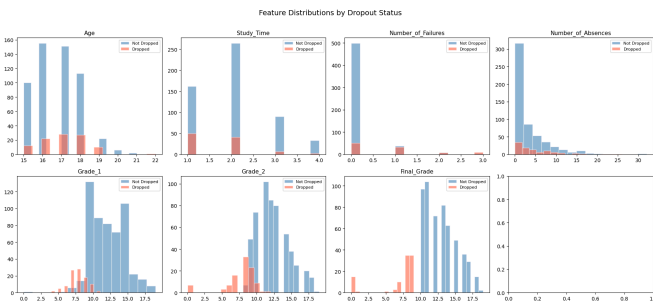


Figure 2: Distribution of key numerical features split by dropout status (blue = not dropped, red = dropped)

c) Grade Comparison

The box plots below clearly show that dropped-out students had significantly lower grades across Grade_1, Grade_2, and Final_Grade. The median final grade for dropout students is noticeably lower, and the spread is wider, suggesting that academic performance is one of the strongest indicators.

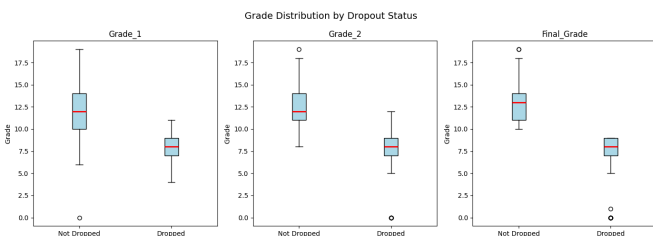


Figure 3: Box plots comparing grade distributions between dropout and non-dropout students

d) Correlation Analysis

We computed Pearson correlations between all features and the target variable. The results show that Final_Grade has the strongest negative correlation with dropout ($r = -0.663$), followed by Grade_2 ($r = -0.592$) and Grade_1 ($r = -0.563$). Number_of_Failures has the strongest positive correlation ($r = 0.380$), meaning students with more failures are more likely to drop out. Wants_Higher_Education also shows a notable negative correlation ($r = -0.310$), suggesting that students who want to pursue higher education are less likely to drop out.

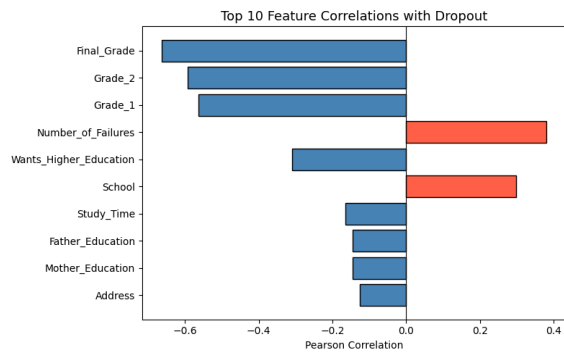


Figure 4: Top 10 features correlated with the dropout target variable

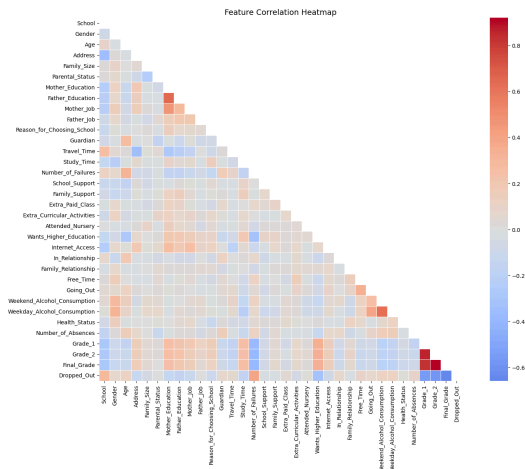


Figure 5: Pairwise correlation heatmap across all features

These findings are consistent with results reported in prior literature. Song et al. (2023) and Cho et al. (2023) both identified academic performance metrics as the strongest predictors of dropout, which aligns with our findings here.

VIII. FEATURE ENGINEERING AND BASELINE REPRODUCTION

A. Engineered Features

Based on the EDA findings, we created four new features to capture patterns that the raw features alone might not represent clearly:

TABLE III

Feature	Formula / Logic	Motivation
Avg_Grade	$(\text{Grade}_1 + \text{Grade}_2 + \text{Final_Grade}) / 3$	Summarizes overall academic performance into a single value
Grade_Improvement	$\text{Grade}_2 - \text{Grade}_1$	Captures whether a student's performance improved or declined over time
Has_Failures	1 if Number_of_Failures > 0, else 0	Binary flag highlighting students with any prior failures, a known risk indicator
High_Absences	1 if absences > median absences, else 0	Binary flag for students with above-average absenteeism

After adding these features, the final feature matrix had 37 columns. Feature importance results from Random Forest confirmed that Avg_Grade was the second most important feature overall, validating the usefulness of this engineered feature.

B. Baseline Model Selection

We selected baseline models that are commonly used in dropout prediction literature. Based on our Phase 2 related work review, the following models appeared most frequently across the five studied papers:

- Logistic Regression: simple and interpretable, used as a standard baseline in most papers
- Decision Tree: used in Song et al. (2023) and Cho et al. (2023)
- K-Nearest Neighbors: distance-based baseline, included for comparison
- Naive Bayes: probabilistic baseline, quick to train
- SVM: used in multiple prior papers as a competitive baseline
- Random Forest: top performer in several related works
- Gradient Boosting: included as an ensemble baseline
- XGBoost: used in Kim et al. (2023) with SMOTE, reported strong results

All models were evaluated both without SMOTE (to see baseline behavior on imbalanced data) and with SMOTE (to see the effect of balancing the training set).

IX. METHODOLOGY, BENCHMARKING, AND EXPERIMENTAL EVALUATION

A. Experimental Setup

All experiments were run in Python 3 using scikit-learn, XGBoost, and imbalanced-learn. The setup was as follows:

- Dataset: 649 students, 37 features after engineering
- Split: 80/20 stratified train/test split (519 train, 130 test)
- Imbalance handling: SMOTE applied to training set only (878 samples after resampling)

- Cross-validation: 5-fold stratified CV on the best model
- Hyperparameters: default settings used for all models in this phase; tuning is planned for the next phase

Each model was evaluated using six metrics: Accuracy, F1-Score, Recall, Precision, and G-Mean.

B. Results Without SMOTE (Baseline)

The table below shows the results when all models were trained on the original imbalanced training set.

TABLE IV

Model	Accuracy	F1-Score	Recall	Precision	G-Mean
Decision Tree	1.0000	1.0000	1.00	1.0000	1.0000
Gradient Boosting	1.0000	1.0000	1.00	1.0000	1.0000
XGBoost	1.0000	1.0000	1.00	1.0000	1.0000
Random Forest	0.9923	0.9744	0.95	1.0000	0.9747
Logistic Regression	0.9538	0.8500	0.85	0.8500	0.9093
SVM	0.9308	0.7273	0.60	0.9231	0.7711
Naive Bayes	0.8923	0.7083	0.85	0.6071	0.8746
K-Nearest Neighbors	0.9231	0.6875	0.55	0.9167	0.7382

Decision Tree, Gradient Boosting, and XGBoost all achieved perfect scores on the test set. While this looks impressive, it is likely caused by overfitting given the small size of the dataset (649 instances). The fact that all three get exactly 1.0 across all metrics, including the 5-fold cross-validation results shown later, is suspicious and suggests the models may have memorized the training data rather than learning generalizable patterns. This will be revisited in the next phase with proper hyperparameter tuning and regularization. Among the remaining models, Logistic Regression performed reasonably well with F1 = 0.85, while KNN struggled the most with recall (0.55).

C. Results With SMOTE

The same eight models were retrained using the SMOTE-balanced training set. Results are shown below.

TABLE V

Model	Accuracy	F1-Score	Recall	Precision	G-Mean
Decision Tree	1.0000	1.0000	1.00	1.0000	1.0000
Gradient Boosting	1.0000	1.0000	1.00	1.0000	1.0000
XGBoost	1.0000	1.0000	1.00	1.0000	1.0000
Random Forest	0.9923	0.9744	0.95	1.0000	0.9747

Logistic Regression	0.9462	0.8372	0.90	0.7826	0.9269
SVM	0.9154	0.7317	0.75	0.7143	0.8421
Naive Bayes	0.8615	0.6400	0.80	0.5333	0.8356
K-Nearest Neighbors	0.8077	0.5902	0.90	0.4390	0.8437

The effect of SMOTE is mixed. For Logistic Regression, recall improved from 0.85 to 0.90, which is the direction we want since identifying at-risk students is the priority. KNN also saw a big recall improvement (0.55 to 0.90) but at the cost of much lower precision (0.9167 to 0.4390), meaning it now flags many non-dropout students incorrectly. The three top models (Decision Tree, Gradient Boosting, XGBoost) remained at perfect scores. The overall takeaway is that SMOTE helps models better detect dropout students, but sometimes at the cost of precision.

D. Model Comparison

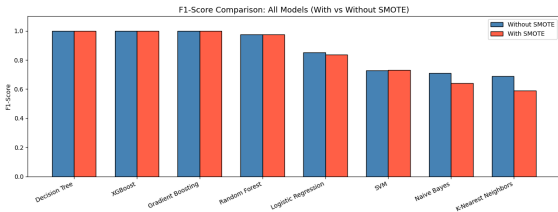


Figure 6: F1-Score comparison for all models with and without SMOTE

E. Best Model Analysis

Based on F1-Score, Decision Tree was selected as the best model (tied with Gradient Boosting and XGBoost). The confusion matrix below shows its predictions on the test set:

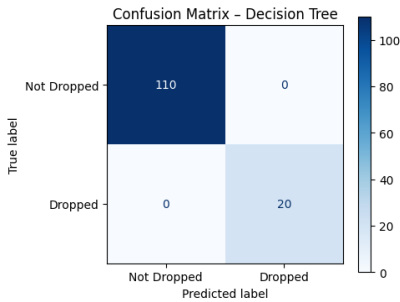


Figure 7: Confusion matrix for Decision Tree on the test set (with SMOTE)

The model correctly classified all 110 non-dropout students and all 20 dropout students with zero errors. Again, we note that this perfect result on a 649-row dataset is a sign of overfitting and should be taken cautiously. The classification report shows precision, recall, and F1 all at 1.00 for both classes.

F. Feature Importance

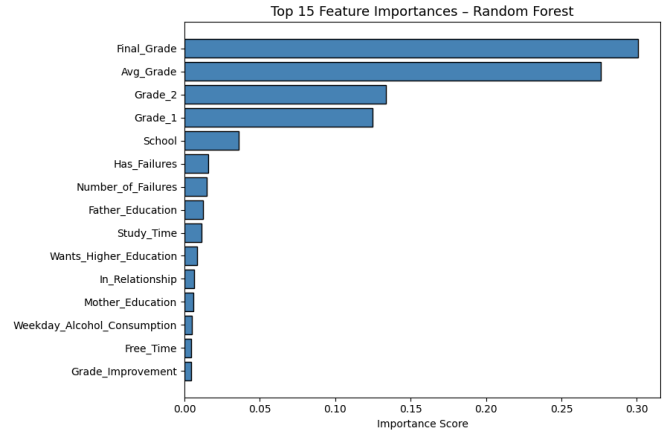


Figure 8: Top 15 feature importances from Random Forest

Random Forest feature importances confirm what we found in the EDA. Final_Grade is the most important feature (0.3006), followed by Avg_Grade (0.2763), which is one of our engineered features. Grade_2 (0.1338) and Grade_1 (0.1246) come next. This confirms that academic performance is the dominant factor in predicting dropout for this dataset. Has_Failures (0.0155) and Number_of_Failures (0.0148) also appear in the top 10, along with Father_Education and Study_Time.

G. Cross-Validation

To check whether the Decision Tree results are stable and not just due to a particular split, we ran 5-fold stratified cross-validation on the full dataset. The results were:

TABLE VI

Metric	Mean	Std Deviation
F1-Score	1.0000	0.0000

The zero standard deviation across all 5 folds further reinforces the concern about overfitting. A model that is perfectly consistent across every fold on a small dataset is very likely memorizing patterns.

X. DISCUSSION

Overall, the experimental results from this phase show that tree-based ensemble models (Random Forest, Gradient Boosting, XGBoost) significantly outperform simpler models like KNN and Naive Bayes on this dataset. This is consistent with findings from Cho et al. (2023), where LightGBM achieved the best F1-score, and Song et al. (2023), where Random Forest and XGBoost were among the top performers.

The main concern from this phase is overfitting in the top three models. The dataset is relatively small (649 rows) and the models may not generalize well to unseen data from different institutions. This is also noted as a limitation in most of the related papers we reviewed. The next steps are:

- Apply hyperparameter tuning (GridSearchCV or RandomizedSearchCV) to find better-regularized configurations
- Test deeper error analysis to understand which student profiles are most likely to be misclassified
- Consider feature selection to reduce noise and improve generalization
- Compare results against the benchmarks reported in prior literature to validate our approach

Despite the overfitting concern, the feature importance and EDA results provide clear and useful insights. Academic grades are the strongest predictors of dropout in this dataset, which makes intuitive sense and is supported by multiple papers in the related work. The engineered feature Avg_Grade being the second most important also validates our feature engineering decisions.

XI. CONCLUSION

This research used a structured educational dataset to provide a machine learning method for predicting student dropout. The findings showed that academic achievement is the most important element influencing dropout prediction through preprocessing, feature engineering, and model validation. When compared to more straightforward techniques, tree-based and ensemble models produced the best results. However, the nearly flawless outcomes seen in a number of models suggest possible overfitting, most likely as a result of the small dataset size. This worry is further supported by cross-validation results, which emphasize the need for better model generalization. In order to reduce overfitting, future research will concentrate on feature selection, hyperparameter modification, and the use of regularization approaches. Furthermore, testing on bigger or external datasets will be necessary to confirm the robustness and

feasibility of the suggested method.

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